Mohd Sarfaraz

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Summary

Computer Science undergraduate (B.Tech, CGPA 8.00/10) seeking Software Engineering, AI/ML or Full-Stack Development internships, with hands-on experience in LLM agents, machine learning, computer vision and full-stack web development.

Education

KIET University — Ghaziabad, India
Bachelor of Technology (B.Tech) in Computer Science; CGPA: 8.00/10 — *Sep 2025 – 2029 (Expected)*

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, DBMS

Experience

AI Research Intern — Dec 2025 – Present
AI Skills Lab, KIET University — *Ghaziabad, India*

- Built **OmniShell**, an LLM agent (TypeScript, LangGraph) that converts natural language into Linux commands.
- Designed self-correcting error recovery; cut command-lookup time by about **70%** over 15+ workflows.
- Prototyped agentic AI workflows (tool calling, multi-step planning, prompt engineering) for lab research projects.

AI Training Contributor (Freelance) — Jul 2026 – Present
Outlier — *Remote*

- Rank LLM responses on coding and reasoning tasks for correctness, instruction-following and AI safety.
- Write reference solutions in Python and C++ used as RLHF and supervised fine-tuning (SFT) training data.

Projects

FreshRoute – Dispatch Optimizer for Perishable Produce
TypeScript, React, HiGHS (WebAssembly), Web Workers, Vite, Vitest, GitHub Actions

- Formulated a mixed-integer programming (MIP) dispatch model solved to a certified optimum in the browser.
- Cut spoilage from **23.3% to 14.7%** vs. FEFO over 11 scenarios; beat both baselines in **30/30** random tests.
- Explained decisions via LP shadow prices; Monte Carlo stress-tested each plan on 100 sampled futures.
- Wrote 57 Vitest tests incl. an objective-vs-ledger check; CI/CD via GitHub Actions to GitHub Pages.

bitandbites Blog – Multi-Tenant AI Blogging SaaS
Next.js, TypeScript, Supabase (PostgreSQL), Claude API, Zod, Tailwind CSS, Docker, Vitest

- Built a SaaS that turns client ideas into fact-checked, SEO-optimized blog posts; onboarded first client.
- Enforced tenant isolation with PostgreSQL row-level security (RLS), verified by a cross-tenant isolation test suite.
- Built a streaming, Zod-validated generative AI pipeline on the Claude API with automated fact-checking.
- Shipped a super-admin console for tenant onboarding and usage tracking; Dockerized, 80+ unit tests.

DepthWizard – Satellite Image to 3D Terrain (Computer Vision) — *SIH 2026 (ISRO)*
Python, PyTorch, Depth Anything V2, FastAPI, scikit-learn, Rasterio, Three.js

- Reconstructed metric 3D terrain from one RGB satellite image (no stereo/LiDAR) via deep learning.
- Built stratified RANSAC calibration to SRTM data: **0.02%** slope error vs. 8.12% for least squares.
- Built a WebGL2 terrain viewer in Three.js with flythrough and live elevation, served via FastAPI.

Technical Skills

Languages: C++, Python, TypeScript, JavaScript, Java, SQL
Web & Backend: React, Next.js, Node.js, FastAPI, Spring Boot, REST APIs, Tailwind CSS, Three.js, WebGL
Databases: PostgreSQL, Supabase, MySQL, Redis, MongoDB
AI / ML: Machine Learning, Deep Learning, Computer Vision, LLMs, Generative AI, AI Agents, RAG, PyTorch
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, CI/CD, GitHub Actions, Linux, Git, WebAssembly
Optimization & Testing: MIP/LP (HiGHS), Monte Carlo, RANSAC, Vitest, Pytest, Unit & Integration Testing
CS Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design

Certifications & Achievements

Certifications: AWS Certified Cloud Practitioner (Amazon Web Services); Linux Fundamentals (Red Hat)
Achievements: Smart India Hackathon (SIH) 2026 – built DepthWizard for an ISRO problem statement